

LUCAS ROMERO

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PROFESSIONAL SUMMARY

Professional Web Developer with experience designing and building responsive, user-friendly websites and web applications. Skilled in HTML, CSS, JavaScript, React, Node.js, and modern deployment tools. Adept at creating clean, efficient code, optimizing performance, and implementing intuitive UI/UX designs. Passionate about delivering scalable web solutions that enhance user experience and drive business growth, with strong attention to detail and problem-solving abilities.

Senior Applied AI/ML Engineer with 8+ years of experience building production systems across regulated enterprise, automotive platforms, and AI-driven products. Proven track record of delivering **real-world ML and GenAI** systems-from data pipelines and ML platforms to RAG and supervised agent workflows-used daily by sales, operations, and analytics teams.

Known for bridging **deep technical execution with business impact**, bringing strong discipline from large enterprises (Pfizer, Toyota North America) and hands-on AI leadership in fast-moving product environments (Relive Realty). Experienced in designing **compliance-aware AI**, scalable ML infrastructure, and GenAI platforms that balance innovation with reliability, security, and governance.

SKILLS

- Applied AI, ML & GenAI

- Machine Learning: XGBoost, LightGBM, scikit-learn, PyTorch
- GenAI: GPT-4o, GPT-3.5, Claude 3, Hugging Face Transformers
- RAG & Retrieval: Pinecone, FAISS, OpenAI embeddings
- Agent & Orchestration: LangGraph, prompt workflows, human-in-the-loop automation
- Model lifecycle: MLflow, offline evaluation, A/B testing

- Data & Platform Engineering

- Data pipelines: Airflow, Kafka, Spark
- APIs & services: FastAPI, REST, event-driven architectures
- Datastores: PostgreSQL, Redis, S3-based data lakes
- Search & ranking: learning-to-rank pipelines, feature engineering

- Cloud & Infrastructure

- Cloud: AWS (ECS, EC2, S3, RDS), Azure (Azure OpenAI)
- Containers & orchestration: Docker, Kubernetes (EKS)
- IaC & CI/CD: Terraform, GitHub Actions, Jenkins
- Observability: Prometheus, Grafana, Sentry

- Security, Compliance & Reliability

- AI governance: access controls, audit logging, privacy reviews
- Compliance-aware AI: fair-housing language filters, PII redaction (Presidio)
- Regulated environments: FDA, HIPAA, GxP development experience
- Production excellence: incident response, reliability engineering, monitoring

WORK HISTORY

Relive

Senior Applied AI/ML Engineer, 01/2024 - 05/2026

- Led the build-out of Relive's first production AI platform, embedding with backend, mobile, product, ops, and compliance teams to deliver ML and GenAI capabilities across the agent ecosystem.
- Implemented property intelligence systems-MLS/partner data pipelines, geo-enrichment, and learning-to-rank models (XGBoost/LightGBM, Python, Pandas)-powering search, recommendations, and curated listing workflows used daily by agents.
- Built ML-driven lead intelligence (conversion scoring, agent-lead matching, follow-up prioritization) integrated into the internal CRM using FastAPI, Redis, and Kafka/SQS, reducing first-response latency by ~25%.
- Established Relive's GenAI stack, integrating GPT-4o and Claude 3 via a secure gateway and delivering RAG pipelines (Pinecone, FAISS, OpenAI embeddings) over SOPs and city playbooks for grounded generation.
- Personally developed the first versions of RAG services and LangGraph workflows, enabling supervised automation for lead intake, call summarization, and task creation that saved agents ~5–7 hours/week in manual admin time.
- Shipped an internal LLM-based Agent Copilot that automated listing descriptions, outbound message drafts, and CRM note generation, adopted by 70%+ of active agents within the first year.
- Designed compliance-aware AI guardrails-retrieval-only fact generation, fair-housing language filters, PII redaction (Presidio), and end-to-end audit logging-supporting AI-assisted workflows with no critical compliance incidents.
- Owned AI service architecture on AWS (ECS/EC2, S3, RDS/Postgres), containerized with Docker, managed via Terraform and GitHub Actions, with observability through Prometheus/Grafana and Sentry.
- Introduced model lifecycle and evaluation practices using MLflow, offline relevance metrics, and A/B testing to tie AI delivery directly to search quality, lead conversion, and agent response time.
- Served as technical lead for AI across multiple product squads, mentoring engineers on ML/GenAI integration patterns and aligning data engineering, backend services, and UX into a cohesive, scalable AI roadmap.

Toyota North America

Senior Software Engineer / Senior ML Platform Engineer, 01/2022 – 12/2023

- Led development of Toyota NA's ML production infrastructure, building end-to-end pipelines for feature engineering, training, deployment, and monitoring using Python, scikit-learn, PyTorch, MLflow, Docker, and Kubernetes (EKS).
- Partnered with data science and security teams to deploy Azure OpenAI-based GenAI pilots, delivering internal tools for report summarization and knowledge-base search using GPT-3.5-Turbo/GPT-4, FAISS, FastAPI, and LangChain.

- Established governance and safety patterns for enterprise AI adoption, including access controls, audit logging, and privacy reviews aligned with Toyota's customer-data and automotive compliance standards.
- Served as technical lead for platform and data engineers, influencing architecture decisions and shaping Toyota NA's modern AI and analytics roadmap.

Data / Platform Engineer, 01/2020 – 12/2021

- Owned batch and streaming data pipelines using Airflow, Kafka, Spark, and AWS S3, enabling advanced analytics across manufacturing quality, supply chain, and connected-vehicle operations.
- Productionized classical ML models-XGBoost, Random Forest, Isolation Forest, and Prophet-for defect prediction, anomaly detection, and demand forecasting, integrating them into data workflows and internal APIs.
- Introduced MLflow for experiment tracking and model versioning, establishing early MLOps practices for analytics and ML teams.

Software Engineer, 09/2018 – 12/2019

- Built backend services and ingestion pipelines for Toyota's connected-vehicle and analytics platforms using Python, Java, Kafka, and AWS (S3, EC2, Lambda), supporting telematics, dealer operations, and internal reporting systems.
- Implemented REST APIs exposing fleet health and operational KPIs, persisting data in PostgreSQL and enterprise data stores.
- Contributed to Toyota NA's early cloud-native modernization, containerizing services with Docker and standardizing CI/CD with Jenkins and Git.

Pfizer

Software Engineer, 07/2016 – 08/2018

- Took ownership of core internal services and data workflows, building and maintaining backend systems that automated clinical-operations tracking and manufacturing quality reporting.
- Designed and implemented REST APIs and integration jobs using Java, Python, and SQL, improving reliability and traceability across multiple enterprise platforms.
- Led development of ETL pipelines that consolidated production, deviation, and quality data into reporting layers used by business and compliance stakeholders.
- Partnered closely with QA, validation, security, and compliance teams to deliver audit-ready systems meeting strict regulatory requirements.
- Contributed to early cloud migration efforts, helping move non-regulated workloads and analytics tools to AWS/Azure pilot environments.
- Became a trusted engineer for incident response, resolving data-flow and system-integration issues under tight operational constraints.
- Mentored junior engineers on enterprise development standards, code quality, and production support practices.

Junior Software Engineer, 06/2015 – 06/2016

- Contributed to internal enterprise applications supporting clinical operations, manufacturing reporting, and quality workflows in a regulated environment.
- Implemented backend features and bug fixes in Java (Spring) and Python, supporting approval workflows, data validation, and reporting tools.
- Built SQL-based reports and ETL scripts to automate manual data pulls from legacy systems into centralized dashboards.
- Gained hands-on experience with FDA, HIPAA, and GxP-compliant development processes, including validation, change control, and audit documentation.
- Supported CI/CD pipelines using Jenkins and Git, assisting senior engineers with releases and production support rotations.